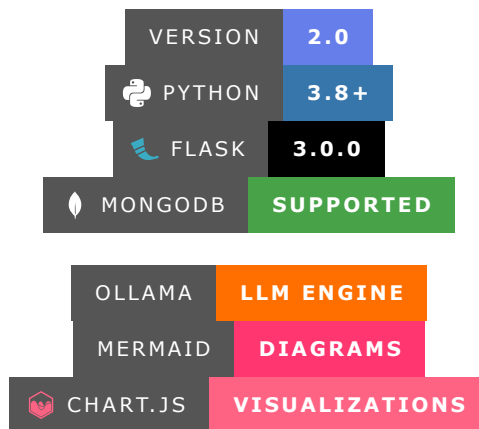




LAPLACIAN

AI Code Assistant & Productivity Platform

by Perfionix AI Technology Pvt Ltd



Company Information

Field	Details
Company	Perfionix AI Technology Pvt Ltd
Product	LAPLACIAN v2.0
Founder	Shubham Rahangdale
Email	connect@perfionixai.com
Status	Production Ready

Executive Summary

LAPLACIAN is an enterprise-grade AI-powered code assistant and productivity platform developed by **Perfionix AI Technology Pvt Ltd**. It combines cutting-edge Large Language Models with modern web technologies to deliver an intelligent, privacy-focused solution for developers, analysts, and professionals.

Platform Highlights

Metric	Value
Total Modules	6 Integrated Modules
AI Models	2 (Core, Coder)
Document Formats	PDF, DOCX, TXT
Data Formats	CSV, XLSX, JSON
Chart Types	4 Auto-Generated
Code Languages	10+ Syntax Highlighted
Diagram Types	15+ Mermaid Diagrams

Key Features Overview

Feature Matrix

Module	Feature	Description
Chat	Multi-Model AI	Switch between Core, Lite, and Coder models
Chat	Web Search	Real-time web search with AI summarization
Chat	Code Execution	Run HTML/CSS/JS in live preview
Chat	Mermaid Diagrams	Auto-render flowcharts, sequences, etc.
Chat	Message Editing	Edit past messages and regenerate

Module	Feature	Description
Chat	Generation Controls	Pause, Continue, Stop AI responses
DocIQ	RAG Q&A	Chat with uploaded documents
VizIQ	Auto Dashboards	Instant KPIs and charts from data
Tasks	Task Management	Priority-based task tracking
Notes	Note Taking	Card-based notes with timestamps
Reminders	Scheduling	Date/time reminder alerts

Chat System - Complete Features

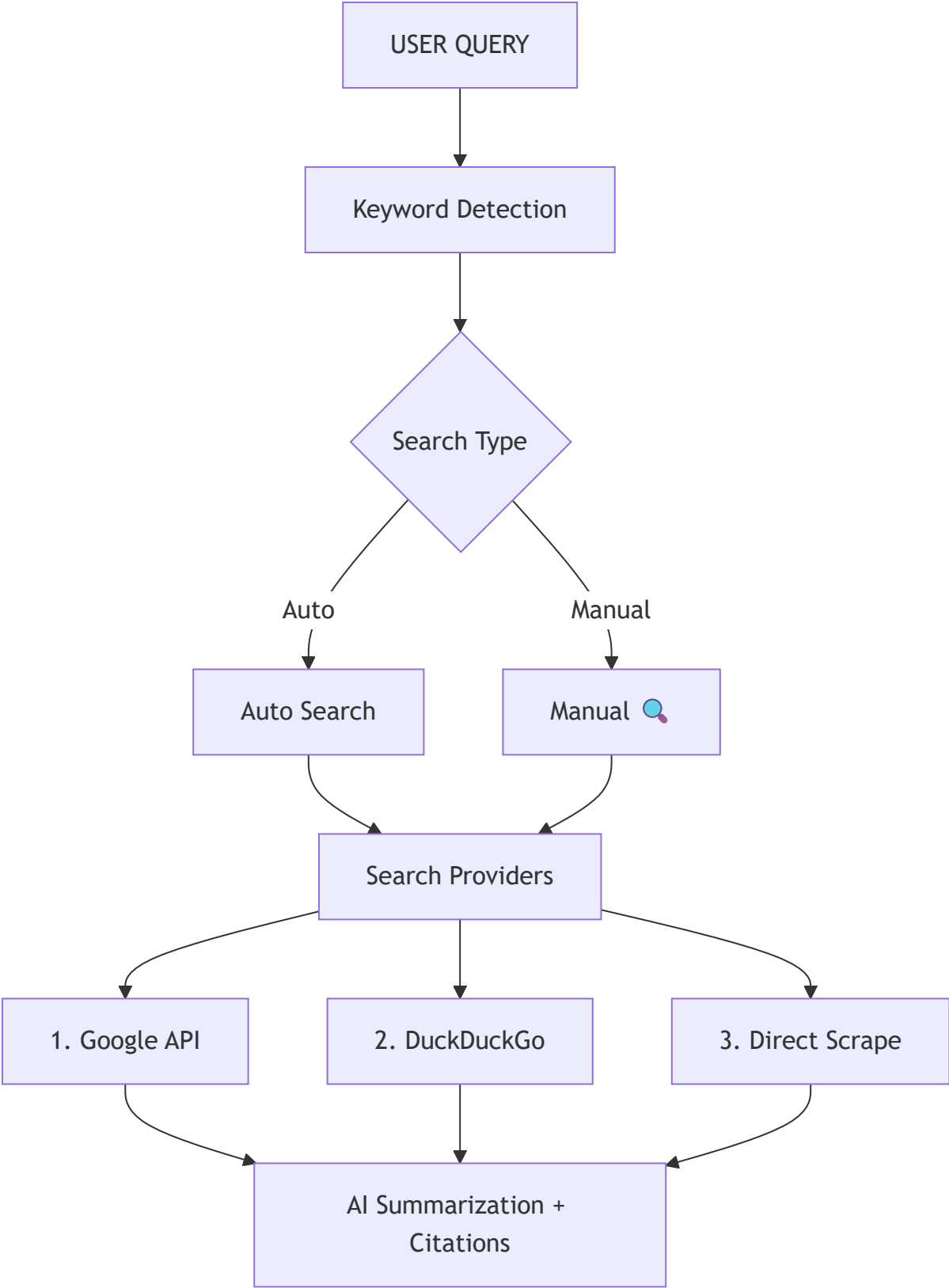
1. Multi-Model AI Selection

Model	ID	Description	Use Case
LAPLACIAN Core	gpt	gpt-oss:20b-cloud	General thinking, complex reasoning
LAPLACIAN Coder	coder	qwen3-coder:480b-cloud	Code generation, debugging

Features:

- Dropdown selector in header
- Real-time model switching
- Availability checking on startup
- Notification on model change

2. Web Search Integration



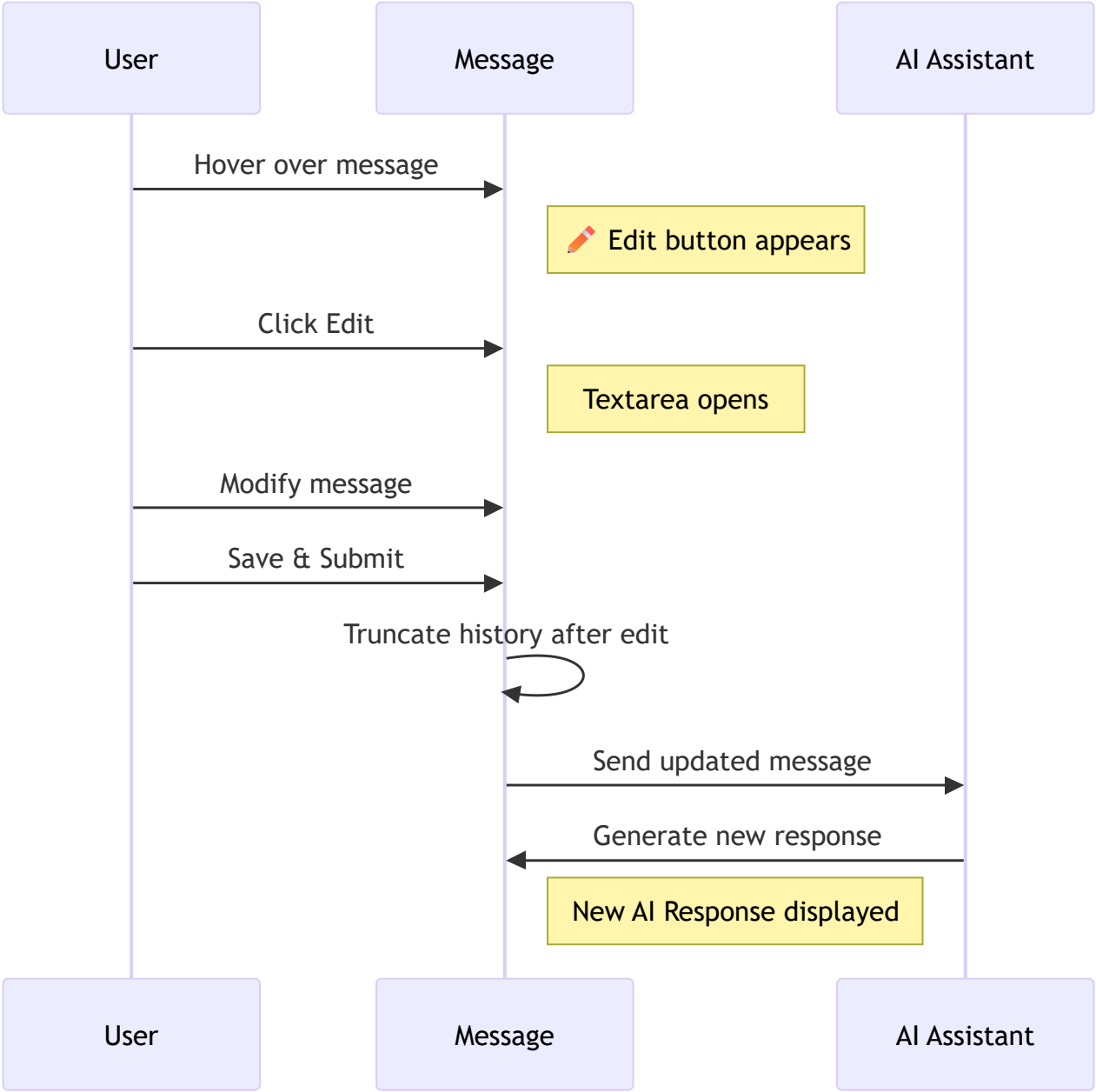
Trigger Keywords

Category	Keywords
Search	search , google , find , look up
Questions	what is , who is , when did
Time	latest , recent , news , current , today
Year	2024 , 2025
Info	price , weather , definition , meaning

Shortcuts

Shortcut	Action
Ctrl + Enter	Force web search
 Button	Manual search trigger

3. Message Editing & Regeneration

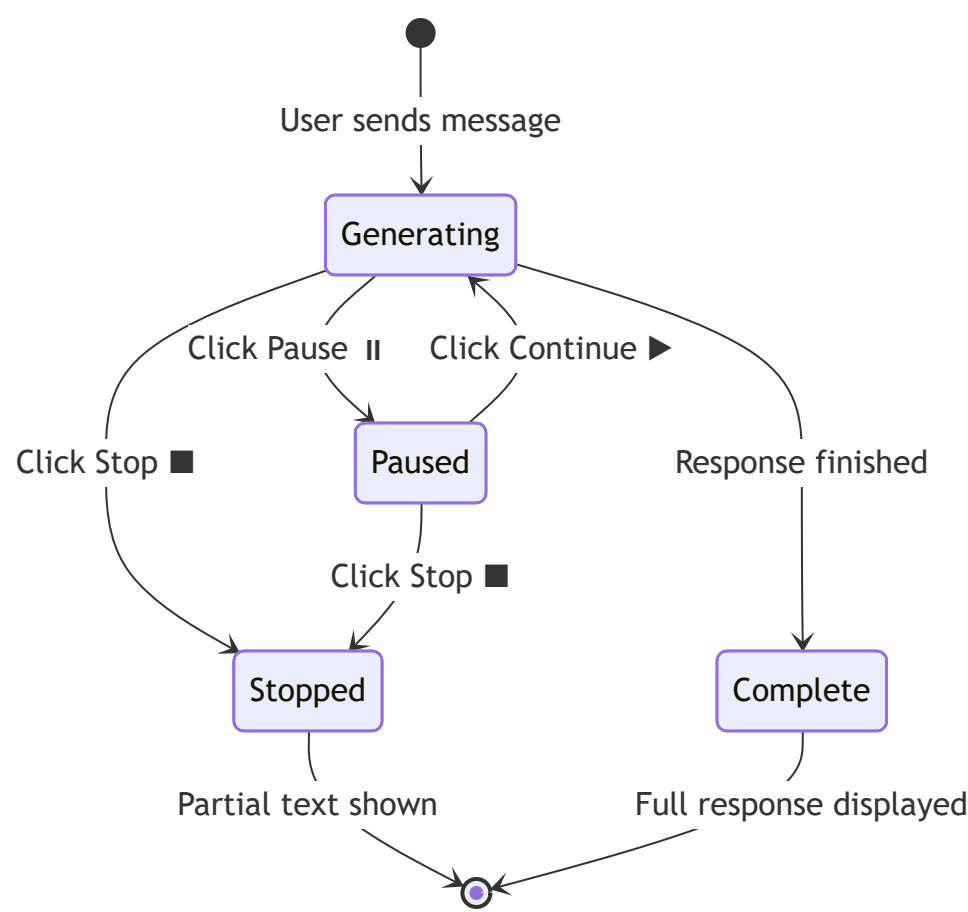


Step	Action	Description
1	Hover	Edit button () appears on user message
2	Click Edit	Textarea opens for editing
3	Submit	History truncated, AI regenerates response

Feature	Description
Edit Button	Appears on hover over user messages

Feature	Description
Inline Editing	Textarea replaces message text
History Truncation	Removes all messages after edited one
Auto-Regenerate	AI generates fresh response
Keyboard Support	Enter = save, Escape = cancel

4. Generation Controls

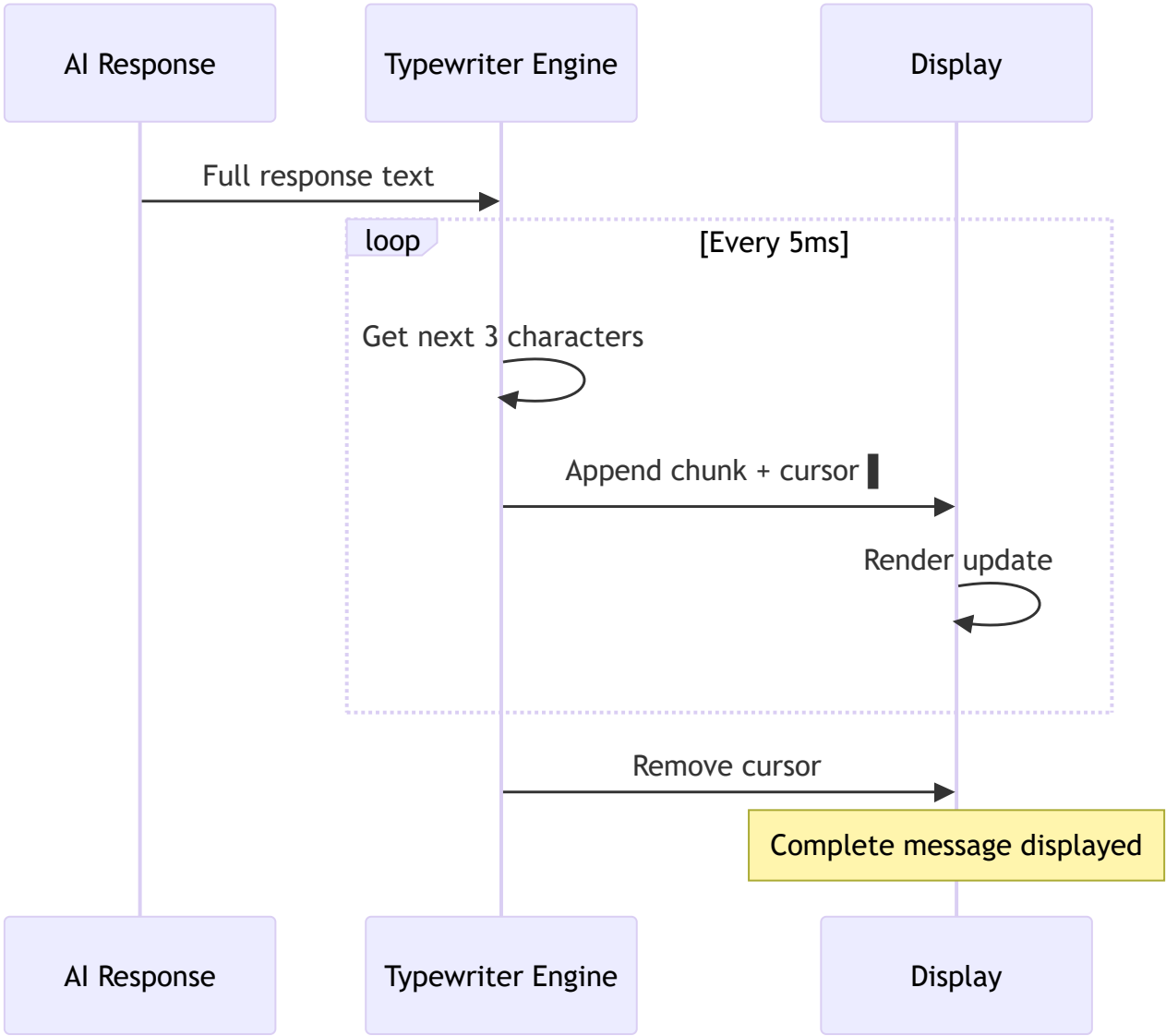


State	Controls Available	Cursor
During Generation	[Pause] [■ Stop]	(animated)
When Paused	[▶ Continue] [■ Stop]	(paused)
When Stopped	Generation stopped message	None

State Variables

Variable	Type	Description
isGenerating	boolean	Currently generating response
isPaused	boolean	Generation is paused
isStopped	boolean	Generation was stopped

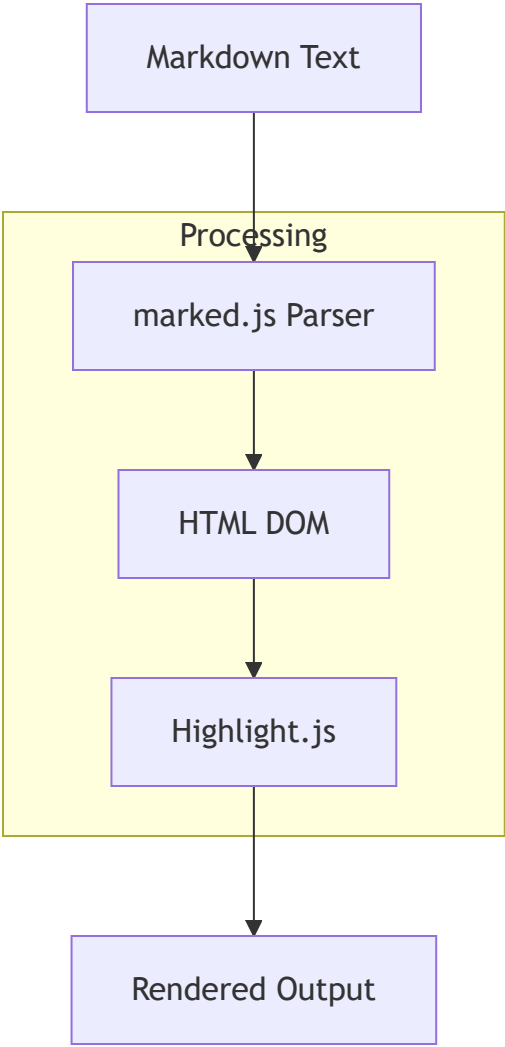
5. Typewriter Effect



Parameter	Value
Speed	5ms per character

Parameter	Value
Chunk Size	3 characters per update
Cursor	█ (animated)
Pause Cursor	

6. Markdown Rendering



Supported Elements

Element	Syntax
Headers	# H1 to ##### H6

Element	Syntax
Bold	**bold**
Italic	<i>*italic*</i>
Strikethrough	~~strike~~
Lists	- or 1.
Code	<code>`inline` or ``` block ```</code>
Links	[text](url)
Images	![alt](url)
Tables	`
Blockquotes	> quote

7. Syntax Highlighting

Supported Languages

Category	Languages
Web	HTML, CSS, JavaScript, TypeScript
Backend	Python, Java, PHP, Ruby
Systems	C++, Go, Rust
Data	SQL, JSON, YAML
Shell	Bash, Markdown

Frontend Code Execution

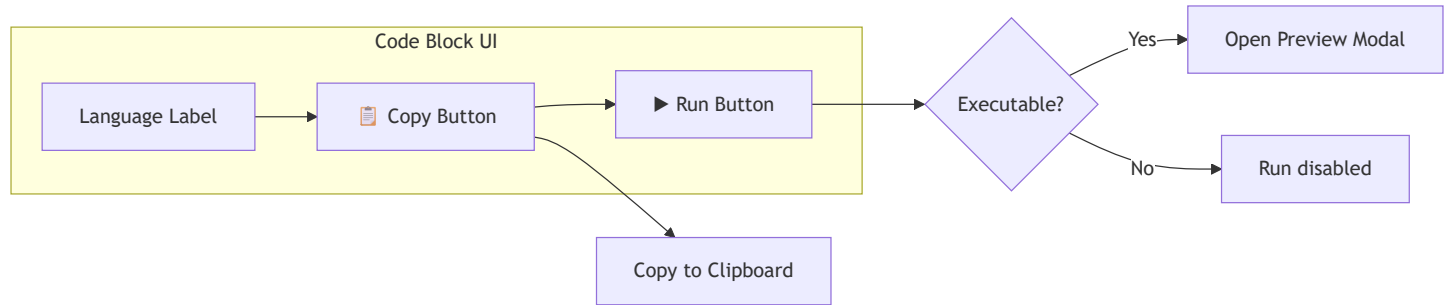
Live Preview Feature

LAPLACIAN can execute HTML, CSS, and JavaScript code directly in a sandboxed iframe preview.

Supported Languages

Language	Execution Type	Description
html	Full document	Complete HTML documents or fragments
css	Demo wrapped	Styles applied to demo elements
javascript / js	Console capture	Output captured and displayed
jsx / tsx	Detected	React/TypeScript components
vue / svelte	Basic	Basic framework rendering

Code Block with Run Button

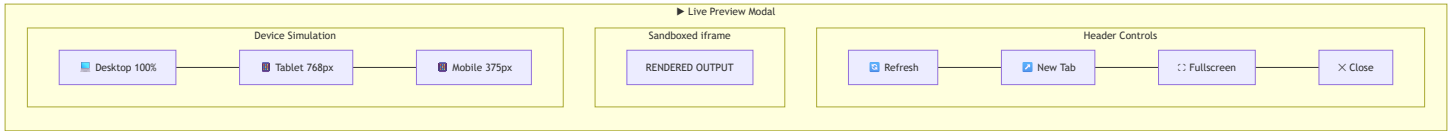


Button	Action
 Copy	Copy code to clipboard
 Run	Execute in preview modal

Detection

```
isExecutableCode(lang, content) {
  const executable = [
    'html', 'css',
    'javascript', 'js',
    'jsx', 'tsx',
    'vue', 'svelte'
  ];
  return executable.includes(lang);
}
```

Preview Modal Features



Button	Function
	Refresh/Reload iframe
	Open in new tab
	Toggle fullscreen
	Close modal

Mode	Width	Icon
Desktop	100%	
Tablet	768px	
Mobile	375px	

JavaScript Console Capture

Method	Color	Example
console.log()	Blue	console.log("Hello");
console.error()	Red	console.error("Error!");
console.warn()	Yellow	console.warn("Warning");
console.info()	Blue	console.info("Info");

Mermaid Diagram Rendering

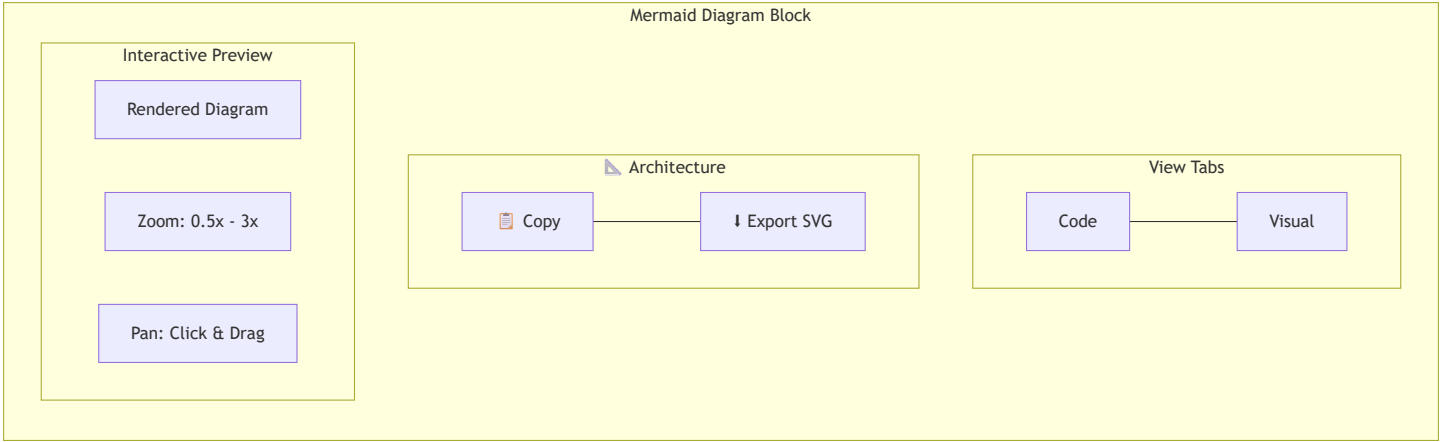
Auto-Detection

LAPLACIAN automatically detects and renders Mermaid diagrams in code blocks.

Supported Diagram Types

Category	Types	Example Syntax
Flow	flowchart , graph	graph LR A --> B --> C
Sequence	sequenceDiagram	A ->> B: Message
Class	classDiagram	class Animal { +name }
State	stateDiagram-v2	[*] --> Active
Entity	erDiagram	USER --o{ ORDER
Gantt	gantt	Task 1: a1, 2024-01-01, 7d
Pie	pie	"A": 30, "B": 70
Other	gitGraph , journey , mindmap , timeline , C4	Various syntaxes

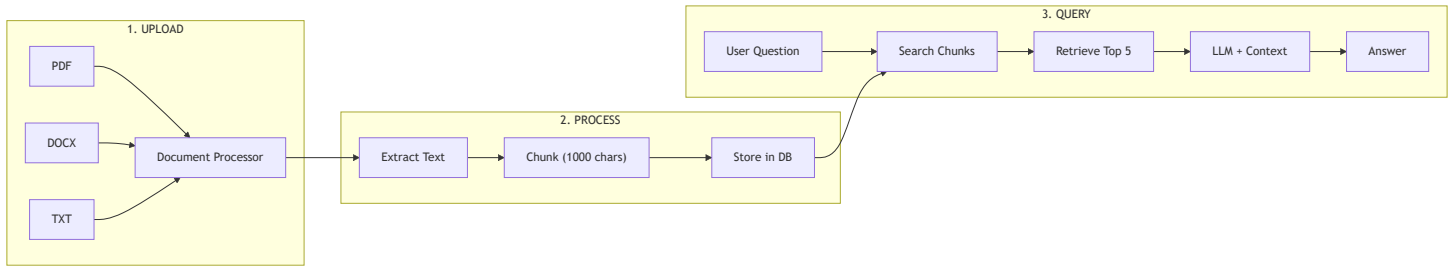
Diagram Block UI



Feature	Description
Tab: Code	View source code
Tab: Visual	View rendered diagram
 Copy	Copy source code
↓ Export	Download as SVG
Zoom	Mouse wheel (0.5x - 3x)
Pan	Click and drag
Fullscreen	Expand view

DocIQ - Document Intelligence

RAG (Retrieval Augmented Generation)



Configuration

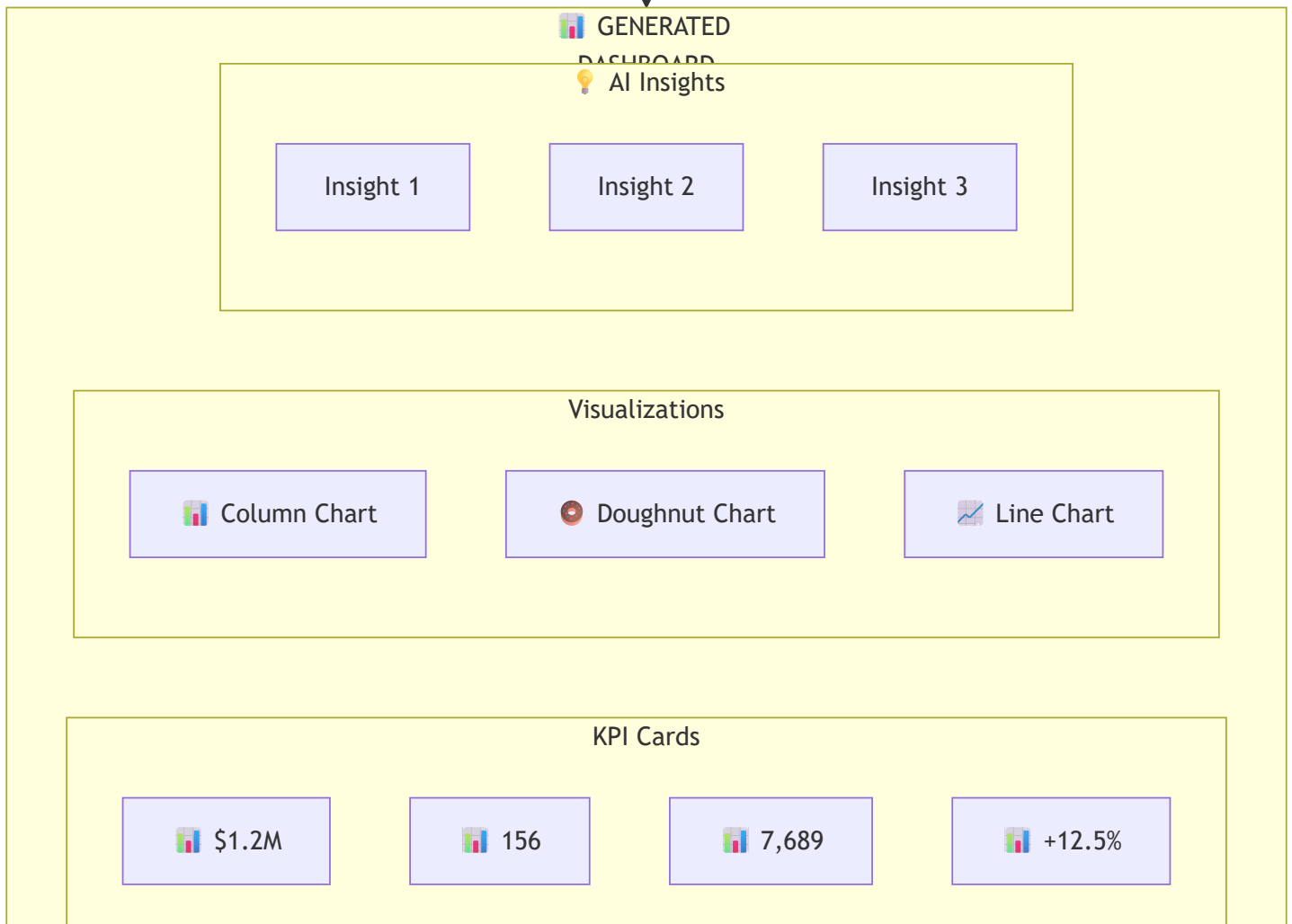
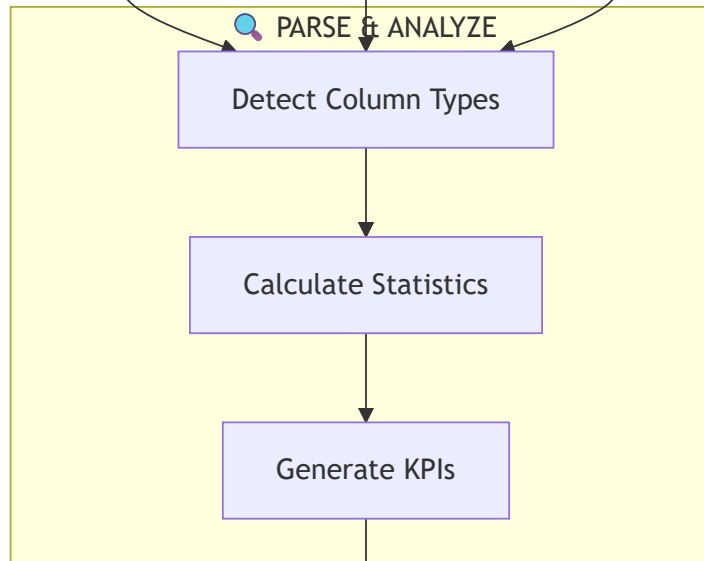
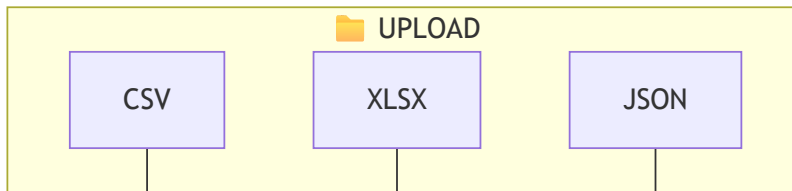
Parameter	Value	Description
Max File Size	16 MB	Maximum upload size
Chunk Size	1000 chars	Text split size
Chunk Overlap	200 chars	Overlap between chunks
Top K Results	5 chunks	Retrieved for context

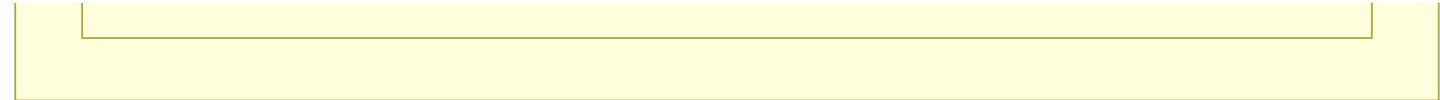
Supported Formats

Format	Library	Features
PDF	PyPDF2	Text extraction from pages
DOCX	python-docx	Paragraph extraction
TXT	Built-in	UTF-8/Latin-1 encoding

VizIQ - Data Visualization

Auto-Dashboard Generation





Generated Components

Component	Icon	Description
KPI Cards		Auto-calculated key metrics
Column Chart		Categorical vs numeric data
Doughnut Chart		Distribution breakdown
Line Chart		Trend analysis over time
Comparison Chart		Multi-metric comparison
AI Insights		Pattern detection alerts
Data Table		First 100 rows preview

Column Type Detection

Type	Detection Rule	Example
numeric	70%+ values are numbers	123 , 45.67
date	Contains date separators	2024-01-15 , 01/15/24
categorical	Everything else	Red , Blue , Active

Productivity Suite

 Reminders Module

Meeting

Dec 25, 10:00 AM

 Notes Module

Note Title

Content...

12:30 PM

 Tasks Module

☐ Task 1 [!]

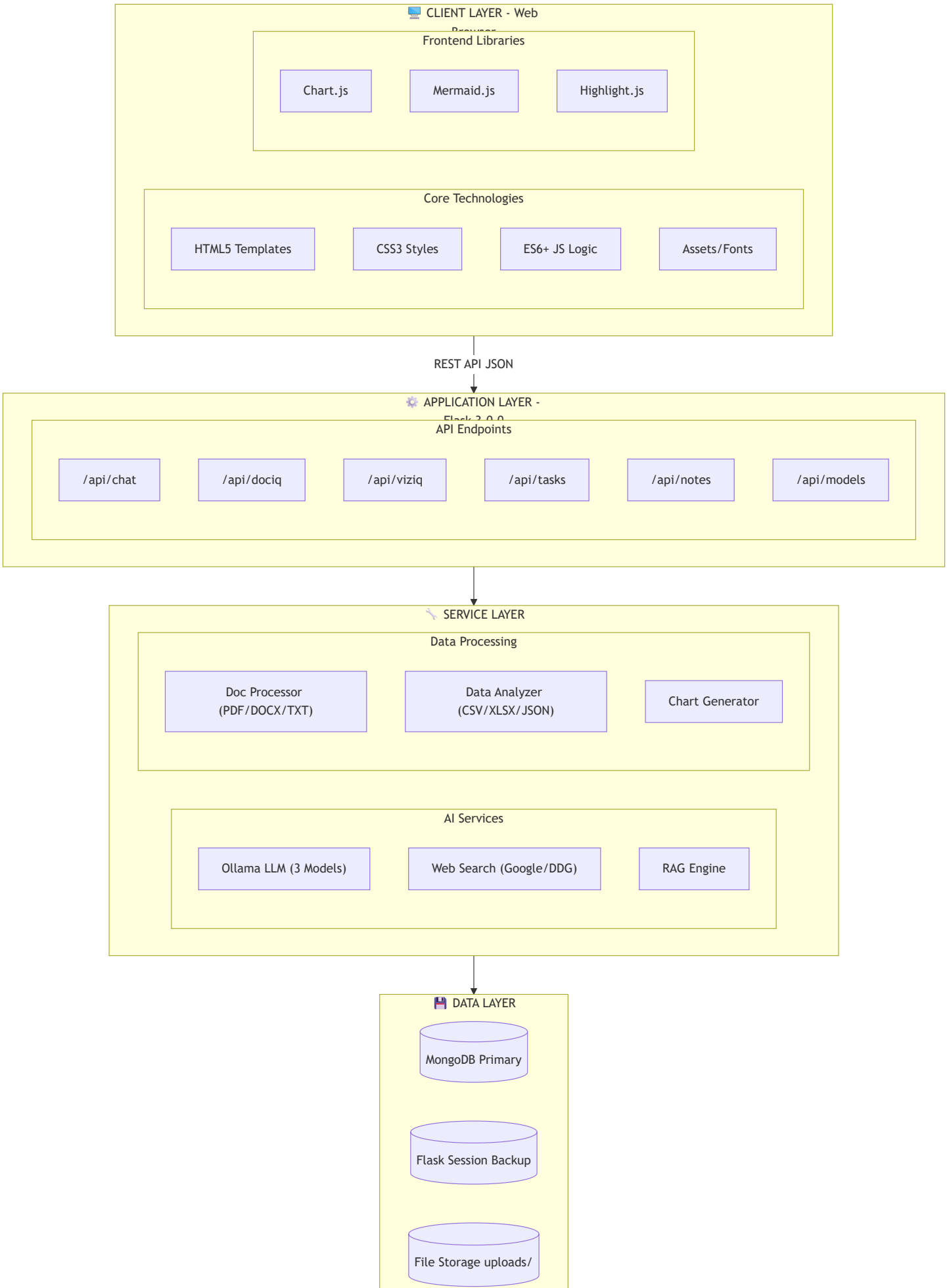
☒ Task 2 [!!]

☐ Task 3 [!!!]

Module	Features
Tasks	Create, Priority (High/Medium/Low), Status (Pending/Done), Filter, Delete
Notes	Create (Title + Content), Card Display, Auto-Timestamp, Delete
Reminders	Create (Date/Time), Alert Notifications, Delete

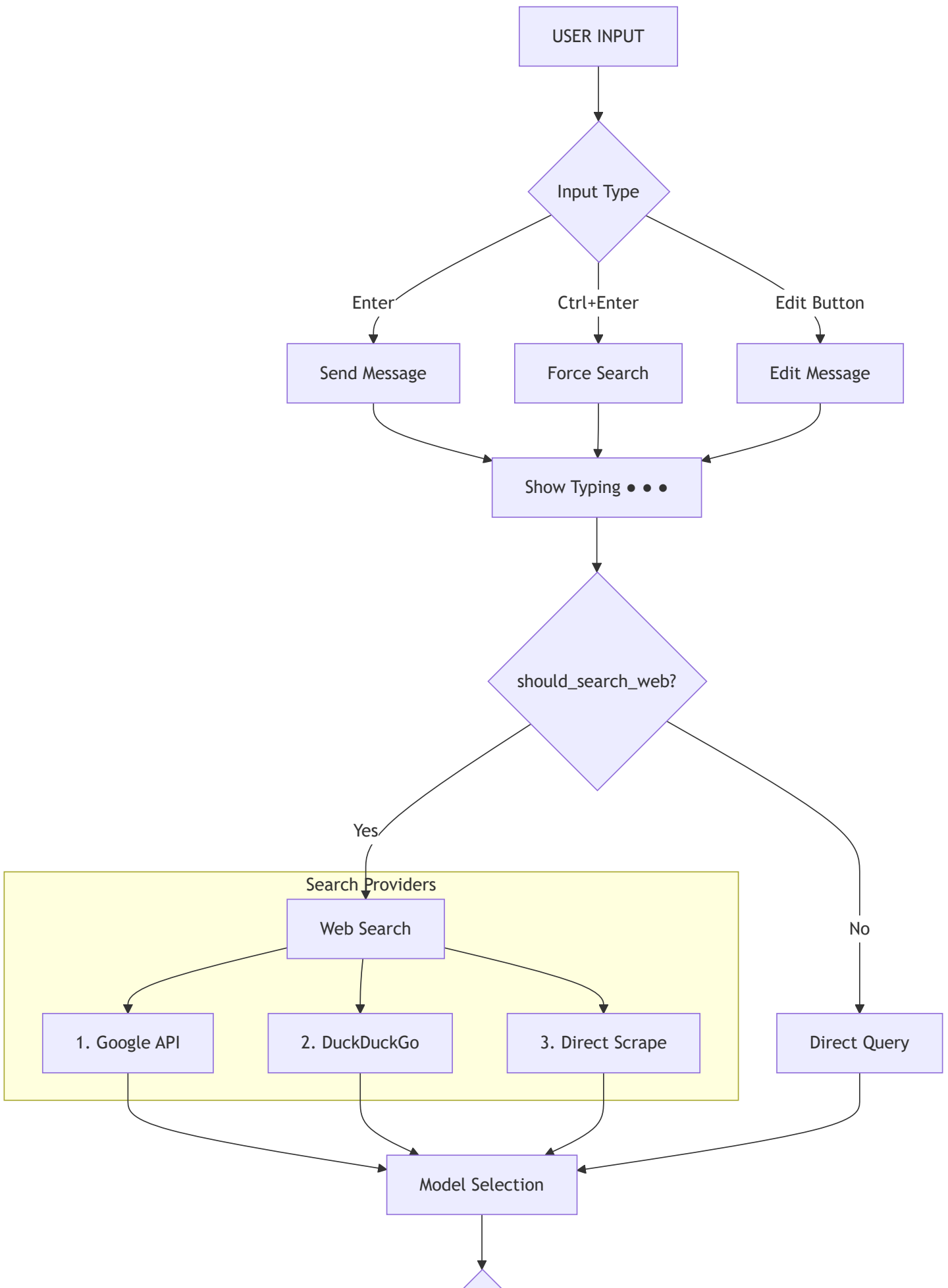
System Architecture

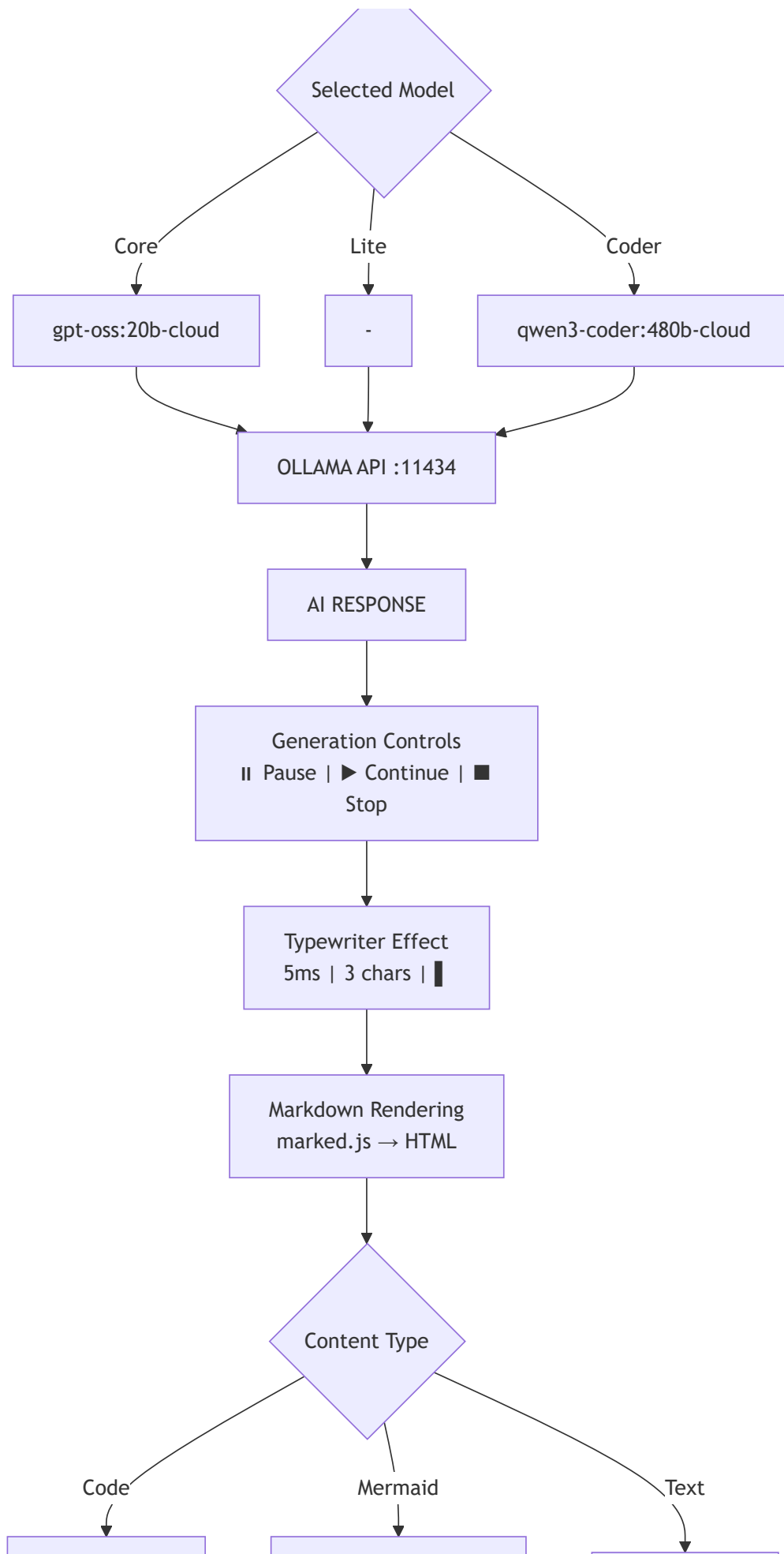
LAPLACIAN Architecture *by Perfionix AI*

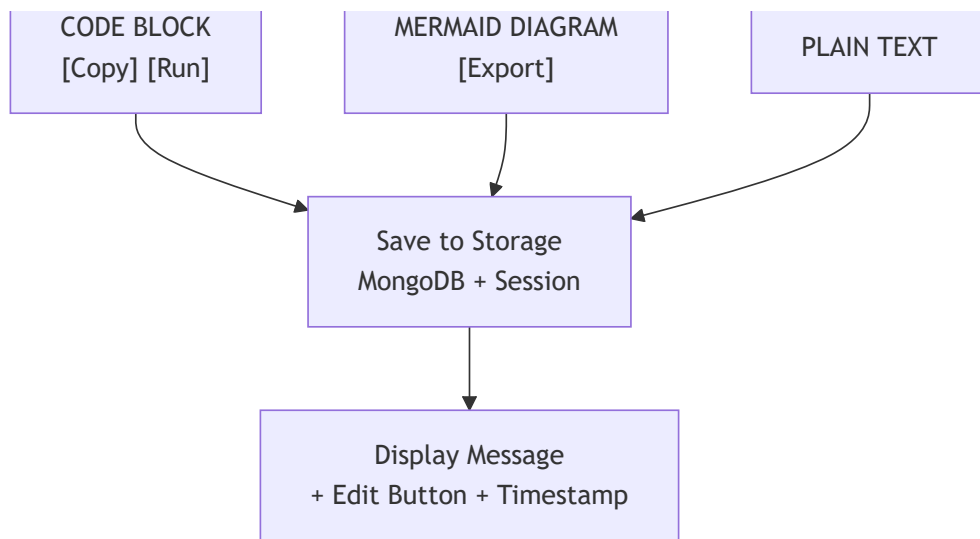


Chat System Flow Diagram

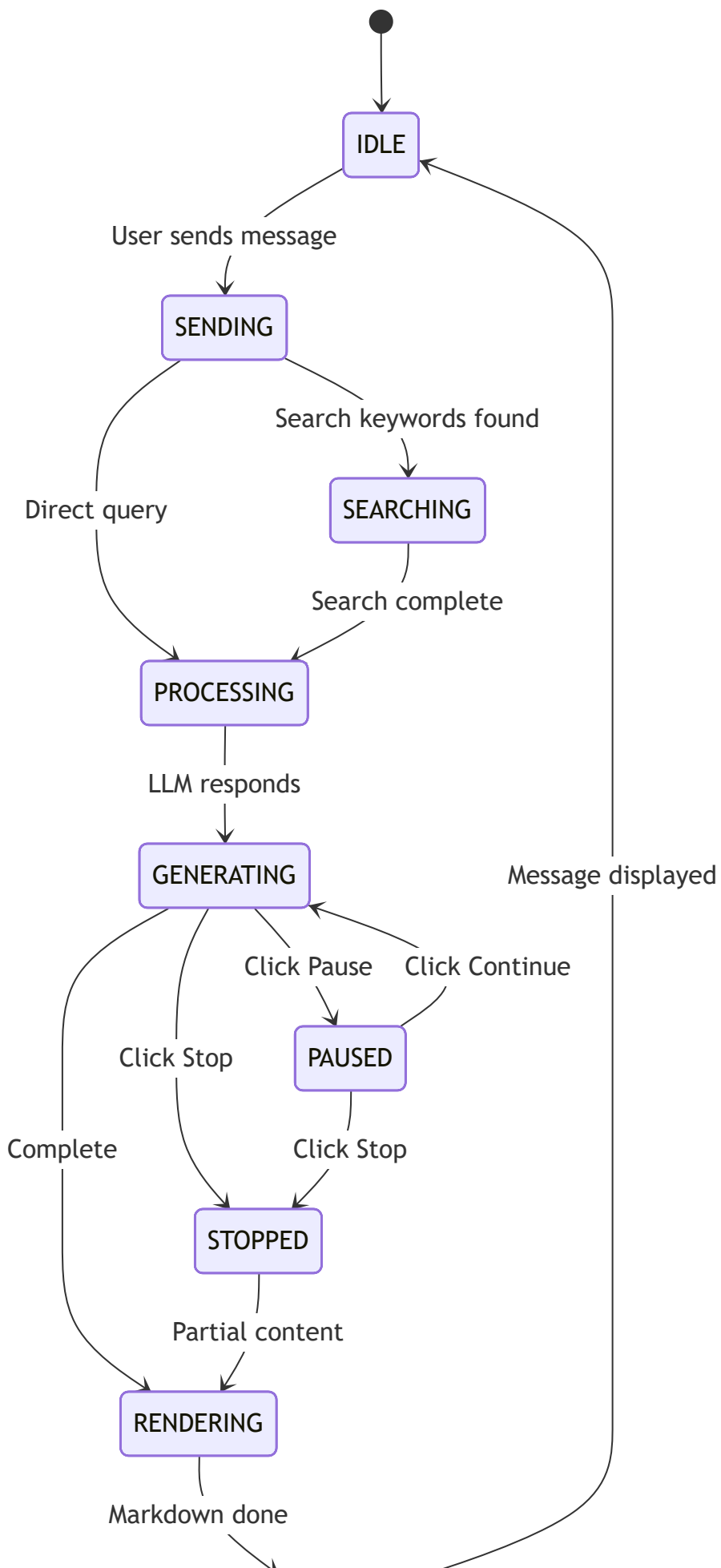
Complete Chat Pipeline - LAPLACIAN CHAT SYSTEM *by*







State Machine





State Transitions

From	To	Trigger
IDLE	SENDING	User sends message
SENDING	SEARCHING	Search keywords found
SENDING	PROCESSING	Direct query
SEARCHING	PROCESSING	Search complete
PROCESSING	GENERATING	LLM responds
GENERATING	PAUSED	User clicks Pause
GENERATING	STOPPED	User clicks Stop
PAUSED	GENERATING	User clicks Continue
PAUSED	STOPPED	User clicks Stop
GENERATING	RENDERING	Complete
STOPPED	RENDERING	Partial content
RENDERING	SAVING	Markdown done
SAVING	IDLE	Message displayed

Tech Stack

Backend

Technology	Version	Purpose
Python	3.8+	Core Language
Flask	3.0.0	Web Framework
pymongo	4.6.1	MongoDB Driver

Technology	Version	Purpose
PyPDF2	3.0.1	PDF Processing
python-docx	1.1.0	Word Documents
openpyxl	3.1.2	Excel Files
BeautifulSoup4	4.12.2	Web Scraping
requests	2.31.0	HTTP Client

Frontend

Technology	Version	Purpose
HTML5	-	Structure
CSS3	-	Glassmorphism Dark Theme
JavaScript	ES6+	Interactivity
Chart.js	Latest	Data Visualization
Mermaid.js	10.6.1	Diagram Rendering
Highlight.js	11.9.0	Syntax Highlighting
Marked.js	Latest	Markdown Parsing

External Services

Service	Purpose
Ollama	Local LLM Inference
MongoDB	Persistent Storage
Google/DuckDuckGo	Web Search
ElevenLabs	Text-to-Speech (Optional)

API Reference

Chat Endpoints

Endpoint	Method	Description
/api/chat	POST	Send message to AI
/api/chat/edit	POST	Edit message and regenerate
/api/chat/reset	POST	Clear conversation
/api/chat/debug	GET	Debug conversation state
/api/models	GET	List available AI models
/api/models/select	POST	Switch AI model
/api/search	POST	Web search with summary
/api/speech	POST	Text-to-speech

DocIQ Endpoints

Endpoint	Method	Description
/api/dociq/upload	POST	Upload document
/api/dociq/documents	GET	List uploaded documents
/api/dociq/documents/<id>	DELETE	Delete document
/api/dociq/chat	POST	Query documents
/api/dociq/summary	GET	Get document summary
/api/dociq/clear	POST	Clear all documents

VizIQ Endpoints

Endpoint	Method	Description
/api/viziq/upload	POST	Upload data file

Endpoint	Method	Description
/api/viziq/data	GET	Get current data
/api/viziq/clear	POST	Clear data

Productivity Endpoints

Endpoint	Method	Description
/api/tasks	GET/POST	List/Create tasks
/api/tasks/<id>	PUT/DELETE	Update/Delete task
/api/notes	GET/POST	List/Create notes
/api/notes/<id>	DELETE	Delete note
/api/reminders	GET/POST	List/Create reminders
/api/reminders/<id>	DELETE	Delete reminder
/api/stats	GET	Get user statistics

Installation Guide

Prerequisites

- Python 3.8+
- MongoDB (optional, recommended)
- Ollama with models installed
- Modern web browser

Quick Start

```
# 1. Clone repository
git clone https://github.com/TR0J49/laplacian-ai.git
cd laplacian

# 2. Create virtual environment
python -m venv venv

# Windows
venv\Scripts\activate

# Linux/Mac
source venv/bin/activate

# 3. Install dependencies
pip install -r requirements.txt

# 4. Configure environment
cp .env.example .env
# Edit .env with your settings

# 5. Start Ollama and pull models
ollama serve
ollama pull phi3:mini
ollama pull qwen3-coder

# 6. Run application
python app.py

# 7. Access Laplacian
# Open http://localhost:5000
```

Future Roadmap

Version 2.1 (Q1 2025)

- ☐ Voice input with speech recognition
- ☐ Multi-language interface support

- ☐ Advanced RAG with embeddings
- ☐ Export chat history as PDF
- ☐ Code execution for Python (sandboxed)

Version 2.2 (Q2 2025)

- ☐ User authentication system
- ☐ Team collaboration features
- ☐ Custom model fine-tuning
- ☐ API rate limiting
- ☐ Webhook integrations

Version 3.0 (Q3 2025)

- ☐ Mobile responsive PWA
- ☐ Plugin architecture
- ☐ Real-time collaboration
- ☐ Enterprise SSO integration
- ☐ Self-hosted deployment guide

Contact & Support

Company	Perfionix AI Technology Pvt Ltd
Product	LAPLACIAN v2.0
Founder	Shubham Rahangdale
Email	connect@perfionixai.com

Technical Support: connect@perfionixai.com
Subject Format: [LAPLACIAN] <Issue Type> - <Brief Description>



LAPLACIAN v2.0

AI Code Assistant & Productivity Platform

Perfionix AI Technology Pvt Ltd

"Empowering developers with intelligent AI"

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